

Alan Braz

Research Software Engineer | AI Systems Architect | Generative AI Manager

Campinas, SP, Brazil (available to relocate to US, sponsorship needed)

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+ Ask anything



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SUMMARY

Senior Software Engineer and AI Engineering Leader with **20+ years** of experience designing and leading enterprise-grade AI, cloud, and software engineering solutions for highly regulated industries. Currently designing and delivering **end-to-end GenAI solutions** that integrate application and solution **architecture** to drive business impact. I bring deep expertise in AI systems (LLMs, RAG, guardrails, and AI agents), hybrid and multi-cloud architectures, and large-scale data platforms, with a strong focus on security, compliance, performance, resilience, and cost efficiency. I lead critical technical decisions, establish reference architectures, and guide best practices for embedding GenAI into production workflows. Previously, I **spent 2.5 years at IBM Research headquarters in Yorktown Heights, NY**, working on cloud-native and microservices architectures, DevOps, and AI-driven software analysis, collaborating with global research and product teams. I hold a **Master's in Computer Science**, have **15 patents and 18 publications**, and a proven track record of leading multidisciplinary teams and scaling research into mission-critical enterprise systems.

CORE EXPERTISE

- AI & GenAI Architecture — RAG pipelines, LLM validation, governance, AI guardrails.
- Enterprise Cloud Architecture — Kubernetes, OpenShift, Multi-Cloud, Terraform.
- Engineering Leadership — Tech strategy, high-performance teams, agile delivery.
- Research to Product — Converting innovation into operational platforms.

PROFESSIONAL EXPERIENCE

IBM Consulting – LA

Lead Generative AI Systems Architect – Data & AI

2025 – Present

- Lead the design and delivery of end-to-end GenAI solutions, integrating solution and application architecture to drive measurable business impact, making technical decisions across LLMs, RAG, and AI agents while ensuring security, compliance, performance, resilience, and cost efficiency, and collaborating with clients, delivery teams, and IBM Technology to build scalable multi-cloud solutions, establish reference architectures, and foster a high-performing regional engineering culture.

IBM Research — Brazil & USA

Research Engineering Manager — AI & Hybrid Cloud

2023 – 2025

- Lead engineering for **LLM systems** including RAG pipelines, guardrails, inference optimization, and enterprise-grade validation frameworks.
- Drive adoption of **Generative AI** to increase engineering throughput across all software delivery phases.

Senior Research Software Engineer & Global Tech Lead

2019 – 2023

- Productized RXN: led end-to-end architecture, DevOps, security and monetization for a multi-tenant SaaS chemistry platform on ROSA (Red **Hat OpenShift Service on AWS**) and converted the Seismic Interpretation Advisor (SIA) research into the commercial GeoScience Advisor (GSA) — cloud-native AI product delivery.
- Cloud & AI engineering lead for Accelerated Discovery: designed and automated multi-cloud infrastructure on IBM Cloud and **AWS with Terraform**, orchestrated **Ray clusters** for model training, and implemented monitoring and cost/resource optimization.
- OSDU Cloud Lead (IBM side): architected distributed Java/Spring Boot APIs on Kubernetes/OpenShift integrating **Microsoft, AWS and Google cloud providers** (leading to a Schlumberger partnership and IBM Open Data for Industries), and created weekly global engineering dev-hours to foster cross-team architecture and code-review culture.

Software Engineering Researcher — IBM TJ Watson (NY HQ)

2017 – 2019

- Built cloud-native front-end and API components using Angular, React, WebComponents.
- Developed ML lifecycle tracking features later incorporated into **IBM Watson Studio**.
- Co-authored and open-sourced **AIMMX**, an AI model metadata extractor published at MSR.

Software Engineering Researcher — Social Data Analytics

2013 – 2016

- Architected real-time Big Data pipelines for social analytics during the **FIFA World Cup**, powering live TV broadcast apps; received **two Research Division Awards**.

IBM Consulting – Intern, IT Specialist, Java Architect, Team Leader

2005 – 2013

- JavaEE systems architecture, Agile coaching, ScrumMaster, DevOps and engineering tooling.

COMMUNITY & LEADERSHIP

- **TechNights Co-Organizer (2023–2025)**: Led 21+ community engineering events with 1200+ participants; topics included AI, Automation, Security, Quantum, Hybrid Cloud.
- **AI Committee Leader (FDC)**: Delivered AI strategy sessions to CXO audiences; organized immersions at IBM Chicago, MIT Watson AI Lab, and IBM NY HQ.
- **TLC-BR Co-Chair**: Guided IBM's 3,000-member technical community to develop leadership and engineering excellence.

- **TDC: The Developer's Conference** (Speaker & Track Coordinator): Coordinated Generative AI tracks (2023–2025) and delivered workshops/talks on prompt engineering, Agentic RAG, GenAI productivity, and related topics. TDC_talks.pdf +1
- **BlueHack**: Cloud Technical Mentor: Served as cloud/AI SME mentoring teams in multiple IBM BlueHack editions (e.g., Atlanta, Jamaica, Yorktown Heights).

PATENTS (Selected, 15 Total)

- Hierarchical Dynamic Deployment of AI Model
- AI model metadata extraction systems (AIMMX)
- Conversational AI evaluation frameworks
- Blockchain-based energy optimization
- Social network-based data recovery

PUBLICATIONS (Selected, 18+)

- “AIMMX: Artificial Intelligence Model Metadata Extractor” – MSR Conference
- “Hybrid Architecture for Multi-Party Conversational Systems” – arXiv
- “AI + Neurologist EEG Annotation Evaluation” – EBioMedicine
- “IBM Design Thinking Software Development Framework” – Brazilian Workshop on Agile Methods

AWARDS

- 2x Research Division Award (RDA) – IBM Research
- IBM Open Innovation Community All-Star
- Best X-Industry App - Watson + GBS Cognitive Challenge

EDUCATION

- Master's in Computer Science (Software Engineering) — UNICAMP
- Bachelor's in Computer Science — UNICAMP

CERTIFICATIONS (Selected)

- GenAI for Business Leaders (Coursera)
- License to Lead – IBM (Leadership Certification)
- IBM Certified Advanced Architect – Cloud v1 and IBM Certified Technical Advocate – Cloud v4
- InstructLab: Democratizing AI Models at Scale (GenAI)
- Sun Certified Enterprise Architect for the Java Platform, Enterprise Edition 5
- Certified ScrumMaster (CSM) - Scrum Alliance

LANGUAGES

Portuguese (Native), English (Fluent), Spanish (Basic)